

Quiet Multicopter Design Rules

What the published record says about building a multicopter that makes less noise: where the rotors go relative to each other and to the arms, how the blades are shaped, when a duct helps, and what linings can and cannot do. Every number here is someone else's, with the conditions it was measured under.

2026-09-28
4 extraction threads + web sweep
~100 PDFs read · 133 abstracts screened
655 quotes verified verbatim
evidence: papers/SoundVisualizer/
design-rules/EXTRACTS-*.md

Read this first. This document is derived from four extracts files in the private papers repo (design-rules/EXTRACTS-LAYOUT.md, -PROPELLER.md, -DUCT-LINING.md, -BROAD.md). Where this page and an extract disagree, **the extract governs**, and where an extract and the paper disagree, the paper governs. Every rule carries its evidence type — **E** experiment, **S** simulation or analytical model, **R** reported second-hand inside another paper, **T** the team's own earlier reviews — and a file and PDF page. Nothing here has been tested on our rig, and our arc is not yet qualified for tones below about 1 kHz (docs/arc-rig-onepager.pdf).

00 The rules, before the evidence

IN SHORT Most of a small multicopter's avoidable noise comes from things passing too close to the blades — an arm, another rotor, a strut, a duct lip — and from spinning faster than the thrust needs. The airframe rules are the cheap ones and the best measured. Duct and lining rules are mostly aerodynamic; the acoustic evidence for ducts is thin and often says they make a static rotor louder.

- Rotor above the arm, arm well below the tip plane.** On a single rotor with a chord-wide arm, moving the arm from 0.1 to 0.2 R below the tip gave the biggest step (up to ~8 dB tonal, ~10 dBA at the lowest microphone); by 0.4 R the tones were back to the isolated rotor [E]. A full-quadcopter simulation put rotors mounted *under* the arms 8.1 dB louder than rotors above them [S].
- In forward flight, keep the rear rotors out of the front wakes.** Raising the rotors 0.33 R above the airframe on standoffs cut a real quadcopter's A-weighted level by 4–8 dBA at –10° pitch [E]. The noise came from the wake the front blade sheds as it passes over its own strut junction, not just from the front rotor.
- Side-by-side spacing has no measured noise rule.** The best available is a simulation: interaction fades out by a tip gap of 0.75–1.0 D, and costs 6–8 % thrust at 0.2–0.3 D [S]. Commercial quadcopters sit at roughly 0.34–0.46 D. Going from two to four rotors on a Phantom added 6–8 dB of broadband where one to two added 3 dB [E].
- Coaxial pairs: axial gap 0.2–0.4 D.** Agreed across several independent experiments; below 0.2 D more blades make it worse [E].
- If you can control rotor phase, do.** Holding two adjacent propellers 90° apart cut the first blade tone by 19 dB, and once phase was controlled the spacing hardly mattered [E].
- Make the thrust at lower tip speed.** Every route that lets the rotor turn slower for the same thrust helps: a twisted blade (+9.3 % figure of merit and up to –4.3 dB [E]), more blades at constant thrust, higher solidity. At equal tip speed, more blades are *louder* in broadband.
- Edge treatments work under conditions, and manufacturing sets the sign.** A boundary-layer trip kills the laminar tones but costs 3–11 % thrust [E]. Trailing-edge serrations need tooth height > 0.25 of the boundary layer, angle < 45° and $St > 1$; 3D-printed cut-in serrations with a blunt root made noise *worse* [E].
- A duct is an aerodynamic device first.** Tip gap dominates its thrust gain, and the literature puts a cliff between about 1 % and 3.6 % of radius [S]. Acoustically, a static 30 cm duct raised noise everywhere on the array, by up to 12 dB, and only helped with 10 m/s inflow [E].
- At drone thickness, only resonant liners reach the blade tone.** No passive absorber can be flat-broadband from ~400 Hz in 10–30 mm [T]; a 30 mm porous layer peaks near 2.8 kHz. Thin tuned liners have cut real fan tones by 3–11 dB(A) [E].
- Judge with what people hear, not dB alone.** A redesigned propeller changed the plain level by under 1.8 dB yet cut modelled psychoacoustic annoyance by up to 20 % [E]. Drones at 5.6 dB lower exposure level were rated as annoying as road vehicles [R].

01 Vehicle layout: arms, spacing, rotor count

Rotor and arm

RULE	NUMBER AND CONDITION	EV.	SOURCE
Keep the arm at least 0.4–0.5 R below the rotor tip plane	Arm below the rotor, width $\approx$ blade chord ($W/c_{0.75R} = 1.08$). 0.1→0.2 R gives the largest drop (up to ~8 dB unweighted, ~10 dBA tonal at the lowest mic); 0.4 R $\approx$ isolated rotor (unweighted); 0.5 R $\approx$ isolated under both weightings. R = 119.2 mm, 5400 rpm, hover.	E	Zawodny-Boyd 2017, p. 7
Mount the rotor above the arm, not below	Under-mounted rotors 8.1 dB louder on average (OASPL over the first 10 BPF harmonics). Phantom-like geometry, hover, 5400 rpm.	S	Zawodny et al. NASA summary, p. 7

RULE	NUMBER AND CONDITION	EV.	SOURCE
Arm shape matters, not only distance	A conical airframe read ~6 dB OASPL above a round rod at the same clearance; the penalty itself depends on clearance (an exception at -0.5 R). No study compares round, rectangular and aerofoil arms.	E	Zawodny-Boyd 2017, p. 8
Raise the rotors off the airframe for forward flight	Standoffs of 32 mm (0.17 R) and 64 mm (0.33 R). Aft rotors elevated: -4 to -8 dBA at -10° pitch, -3.5 to -5 dBA at -4°. Hover: single rotor -1 to -4 dB narrowband; full vehicle inconclusive (-3.0 to +1.3 dB). Standoffs added tones at 1.5 and 4.5 BPF from structural vibration.	E	Zawodny-Pettingill-Thurman 2022, pp. 1, 4, 6-8, 11
A close arm also acts as a small ground plane	All rotor-airframe cases showed a slight thrust <i>increase</i> over the isolated rotor.	E	Zawodny-Boyd 2017, p. 9
Installation can outweigh the rotor's own loading	Strut installation and forward flight at negative pitch generate far more sound than the steady thrust loading of the blades (dipole source-mode model).	S	Roger & Moreau 2020, p. 1
A body upstream of the rotor imprints its own tones	Rotor behind a body of revolution: humps at BPF multiples from blade-to-blade correlated boundary-layer turbulence (LES). A rotor partly immersed in a wall boundary layer shows the same "haystacking" (experiment).	S/E	Zhou 2024, p. 1; Morton 2012, p. 11

"Elevation of the rotors from the vehicle airframe yielded nearly 8 dBA overall noise reduction in forward flight, while yielding up to 4 dB reduction in overall tonal levels for one of the rotors in hover."

Zawodny, Pettingill & Thurman, Inter-Noise 2022 (NASA NTRS 20220006250), p. 1

Rotor to rotor

RULE	NUMBER AND CONDITION	EV.	SOURCE
Side-by-side tip gap: interaction fades by 0.75-1.0 D	Thrust -7.93 % at 0.2 D, -5.91 % at 0.3 D; oscillations mostly gone at 0.75-1.0 D; OASPL at 0.2 D much higher than at 1.0 D. Four isolated 9.45-in rotors, no arms or body in the sweep.	S	Lee & Lee 2020, pp. 16-18, 28
What real vehicles do	DJI Phantom 2: 0.35 m diagonal for 0.24 m rotors (≈ 0.46 D tip gap). NASA SUI Endurance: 510 mm hub-to-hub, R = 190.5 mm (≈ 0.34 D). Tip gaps are our arithmetic.	-	Lee 2020 p. 11; Pettingill 2022 p. 3
More rotors add noise non-linearly	Broadband +3 dB going 1→2 rotors, +6-8 dB going 2→4; tones +3-8 dB. A smaller propeller on the same frame (bigger gap) gave only 3-5 dB for 2→4. DJI Phantom II, static.	E	Intaratep 2016, pp. 4-5
Coaxial axial gap $z/D = 0.2-0.4$	Minimum noise and a tonal/broadband balance at 0.2-0.4 D; loudness minimum at 0.3-0.4 D. Best at 8 N: z/D 0.4 with 4 blades (on-axis OASPL), 0.3 with 5 blades (sound power). Below 0.2 D, more blades raise tonal noise. One cited listening study found 0.1 D best for a 14-in rotor, so the optimum may shift with size.	E	Hirono 2024, pp. 1-2, 19
Very close coaxial pairs are loud	15-in contra-rotating pair: 17 mm spacing gave significantly higher OASPL than 48 mm; OASPL falls with spacing up to 70 mm, broadband flat (79-82 dB).	E	McKay 2021, pp. 7, 14, 16
Coaxial is the loudest octocopter layout	Peak SPL 93 dB, 5 dB above the single-plane layout and 3 dB above the hybrid one.	S	Aziz & Shi 2024, p. 1
Rotor phase is a separate, larger lever	Two 9-in props at s/D 1.01-1.05: -19 dB at the first BPF at 90° (2 blades), -15 dB at 36° (5 blades); in-phase is worst for level and annoyance; spacing had minimal impact once synchronised. Needs active phase locking.	E	Turhan 2025, pp. 1, 5, 8
... and it redirects rather than removes	Quarter-period offset between two hovering rotors: up to -5 dB(A) directly beneath the pair. Inflow does not move the phase that gives the minimum.	S/E	Chai 2025, p. 12; Del Duchetto 2025, p. 1

"with optimal axial spacing - between $z/D = 0.2$ and 0.4 - showing minimum radiated noise levels, and a general balance between tonal and broadband noise."

Hirono, Robertson & Torija, J. Sound Vib. 569 (2024) 117971, p. 19

Ground

RULE	NUMBER AND CONDITION	EV.	SOURCE
Within one diameter of the ground, expect ~6 dB more	In ground effect when $L \leq D$. Any ground plane raises OASPL in the reflection zone by roughly 6 dB; extreme IGE adds another 1-2 dB. Single 10×5.5-in propeller.	E	Hanson 2023, pp. 2, 11
A porous ground helps only partly	Solid plate +3 dBA at shallow angles at $L/R = 1$; porous treatment removed about 2 dBA of it.	E	Jawahar 2025, p. 6

02 Propellers and blades

RULE	NUMBER AND CONDITION	EV.	SOURCE
Buy the thrust with efficiency, then drop rpm	Lower rpm lowers tip Mach, drops the tone frequency and weakens turbulence ingestion at once. Chord + twist + aerofoil optimisation of a 2 N MAV rotor: the final rotor radiates 10 dB(A) less and uses 4 W less at the same thrust. The model over-predicted the tonal gain at BPF (25 vs 15 dB(A) measured).	E+S	Serré 2019, pp. 10–12 and Fig. 18
Twist the blade	+9.3 % figure of merit and up to −4.3 dB OASPL at the same thrust and diameter; the untwisted blade adds broadband at 2–10 kHz. 200 mm, 5400 rpm. Twist and camber changed together.	E+S	Sun et al. 2023 (Drones), pp. 1, 10–11
More blades at constant thrust; not at constant tip speed	Constant thrust: more blades need lower rpm and give lower broadband. Constant tip speed: more blades give <i>more</i> broadband (5 blades ~5 dB above 2 upstream). 9-in, solidity not held constant.	E	Baskaran 2024, pp. 1, 7, 9, 16
... but the gain is in the rotor plane	Iso-thrust 2→5 blades: OASPL −10 dB in the disc plane, nearly constant at ±60° because broadband rises. Aerodynamic optimum at 3 blades (a reproduced vortex-lattice prediction).	E+S	Gojon 2021, pp. 3824–3825
Higher solidity quiets through lower rpm	At equal rpm similar propellers give the same harmonic levels; the higher-solidity blade was quieter at equal thrust only because it spun slower.	E	Jordan 2020, pp. 14, 18
Trip the boundary layer (and pay for it)	Removes laminar vortex-shedding tones. Broadband −8.87 dB (optimised rotor) / −3.87 dB (commercial) out of plane; up to −21 dB at 6.1 kHz at low tip speed. Thrust −3 to −11 %. Tripping the outer 50 % of span gave the full benefit. R = 190.5 mm. A separate study: −5 dBA isolated, only −1.5 dBA on the full hovering drone.	E	Pettingill-Zawodny 2026 TM, pp. 27, 41; Zhao 2023, p. 2
Serrations only work inside three limits	Tooth height $h^* > 0.25$, included angle $< 45^\circ$, $St_6 > 1$; tooth count does not matter. CNC-machined: −7.5 dB in hover, −12 dB forward, ~5 % lift loss forward.	E	Candeloro 2020, pp. 5, 10; Li 2018, pp. 1, 4, 6
How the serration is made decides its sign	Printed cut-in serrations with a blunt ~1.4 mm root <i>raise</i> broadband (root vortex shedding at $St \approx 0.12$ on root thickness); add-on serrations reduce it. Tripped add-on serrations cost up to 9.5 % figure of merit. Nominally identical printed samples differed by 2–4 % aerodynamically.	E	Santamaria 2026, pp. 5, 13, 16, 20
A plain trailing-edge plate can make it louder	Plate attachment: OASPL > 3 dB higher, drag +25 %. Serrated Gurney flap: optimum 3 mm height, 1 mm gap, 3–6 cm from the tip, within a 10 % efficiency budget; 200–800 Hz rises in every variant.	E+S	Noda 2022, pp. 6, 8–10
Uneven blade spacing trades tone for sub-harmonic	Up to −5 dB A-weighted tonal near the disc plane, up to +4 dB below 50°; energy moves to 0.5 BPF, where A-weighting hides it. Clustered pairs cost 8.5 % thrust and add broadband unless the pairs are axially separated. An optimisation found blade sweep more robust than uneven spacing.	E, S	Kim 2016 thesis, pp. 39–56; Fruncillo 2025, p. 1
Toroidal propellers are not reliably quieter	Careful test of a commercial 5-in toroidal: no noise or efficiency advantage, up to 10 dB louder than a high-pitch conventional prop (which was the least efficient), excess above 3–4 kHz. Other studies report 5–20 dB reductions; the spread tracks method rigour.	E	Meister 2026, pp. 1, 7, 9; Wei 2024 (Drones), p. 1
Loop-type blades help in a narrower way	Joining adjacent tips: −4.7 dB OASPL isolated, −3.5 dB installed, lower loudness and tonality, slightly lower figure of merit, more mass. 240 mm.	E	Sun et al. 2023 (IJERPH), pp. 16, 20–21
Material and print route are noise variables	A 254 mm propeller printed in clear resin was quieter than wood and aluminium copies at similar figure of merit. Roughness from print settings did not map cleanly to noise.	E	Chen 2025, p. 10; Popister 2025, p. 10
Combined edge devices can beat single ones	Winglets + serrations: +9.7 % thrust, −2.7 dB low-frequency broadband, up to −9.1 dB on some BPF tones. Owl-serration + cicada geometry: −5.5 dB vs an industry benchmark blade.	E	Khalaf 2025, p. 1; Wei 2024 (Nat. Comm.), p. 4

“boundary layer tripping reduces thrust by 3–11% depending on the rotor type, with the OPT2 rotors showing more sensitivity to tripping than COTS rotors.”

Pettingill & Zawodny, NASA/TM-20250003316 (2026), p. 41

“how the serrations are manufactured is crucial and must be assessed during the design phase.”

Santamaria et al., J. Sound Vib. 626 (2026) 119656, printed p. 16

03 Ducts and shrouds

RULE	NUMBER AND CONDITION	EV.	SOURCE
Tip gap is the parameter every source agrees on	5-in 3-blade, CFD: 0.6 % and 1.1 % of radius comparable, sudden drop at 3.6 %; FM* from ~30 % open to >60 % at 0.6 %. Design study: 3.04→1.71 % R gave +17.85 % hover efficiency; 3.04→5.17 % R cost up to 18.1 %, below an open rotor.	S	Chew-Gan-Hesse 2021, p. 14; Dayhoum 2025 (ApplSci), p. 9
... the first widening hurts most, and the size scales it	0.5→1 % D_t cut FM 24 % (0.67→0.51). 0→4.4 % D_t cut thrust ~84 % on a 48-in model but a similar range cost only ~20 % on the 7-ft X-22A fan. No single percentage transfers across sizes.	R	Pereira 2008, pp. 70, 74, 101
Round the inlet lip generously	Lip radius 1.4→8.3 % D_t ; FM 0.72→0.89 with no diffuser [R]. Recommended 3–15 % of rotor diameter [S]. A critical 6 % has been proposed, but separation was seen at 12.5 % too [R].	R/S	Pereira 2008, p. 70; Dayhoum 2025, p. 9; Zhang-Barakos 2020, p. 23
Inlet separation, not tip gap, drives a duct's broadband penalty	Hubbard's shrouded propeller: with attached flow, peak pressure about half the open propeller's; with separation at the lip, almost twice. A too-small lip radius raises broadband; tip clearance was reported acoustically insignificant (10-in fan).	R	Zhang-Barakos 2020, p. 29; Simon 2022, p. 1
Diffuser: expansion ratio matters, angle and length less	Classical optimum 10° and 50 % of throat diameter [E]; exit area ratio 1.2–1.3 gave up to 15 % more static thrust [R]; length is second-order (–3 % thrust across 18–144 % R) [S]; one CFD sweep found no correlation at all [S]. Three historical studies disagree on the sign of the angle effect.	E/R/S	Pereira 2008, pp. 1, 77; Dayhoum 2025, p. 9; Ma'arof 2023, p. 314
Centre the propeller axially for tone attenuation	Best attenuation with the propeller centred in the duct, from interference between upstream and downstream waves. NASA's 10-in duct put it at 45 % of chord behind the leading edge. Forward position is better in axial flight, rearward in static.	E/R	Simon 2022, p. 1; Zawodny 2025 TM, p. 9; Pereira 2008, p. 78
Duct thickness at the prop plane ≥ 10 % of D	To avoid separation at static conditions (Black & Wainauski, as cited). Area ratio 1.15, tip gap 0.05 in (1.27 mm) in the NASA design.	R	Zawodny 2025 TM, pp. 8–9
Keep struts away from the rotor plane	Moving the struts 5.5 % D_t further from the rotor cut noise by up to 10 dB, back to the no-strut level (fenestron). Stator vanes that double as struts recover swirl: +11 % to +26 % thrust.	R	Pereira 2008, pp. 88, 119
A static duct can be louder everywhere	30 cm duct, 2 mm tip gap, no inflow: "no regions with reduced levels", up to +12 dB at the array's top and bottom; broadband up, first harmonics down. With 10 m/s inflow the same duct reduced noise at every microphone.	E	Malgoezar 2019, pp. 6, 9–10, 16
Compare ducted and open at equal thrust	The duct changes the loading: NASA needed +4.6 % rpm in hover (+16.5 % in axial inflow) to match thrust. Part of an equal-rpm "reduction" is lost thrust.	E	Zawodny 2025 TM, p. 49; Malgoezar 2019, p. 15
Per-propeller shrouds beat one whole-frame duct, aerodynamically	+15.5 % lift / +24.0 % FM vs +7.0 % / +9.7 % on the same quadrotor. CFD, no acoustics. Nobody has measured both aero and noise on the same airframe.	S/T	Li-Yonezawa-Liu 2021, p. 22; Koszałka ducting review, pp. 2, 5
For a fixed size limit, a bigger open rotor may win	"a larger-diameter open rotor is more likely to give better performance than a smaller-diameter shrouded rotor", ignoring safety benefits.	E	Pereira 2008, p. 2

"While the tip clearances of 0.6% and 1.1% produce comparable results, a sudden drop in performance can be observed for 3.6% tip clearance."

Chew, Gan & Hesse, AIAA SciTech 2021 (doi:10.2514/6.2021-0212), p. 14 – CFD, 5-in propeller

"No regions with reduced levels of noise are present and an increase of noise up to 12 dB is found at the top and bottom of the array."

Malgoezar et al., Int. J. Aeroacoustics 18 (2019), p. 9 – static case, no inflow

04 Linings and materials

RULE	NUMBER AND CONDITION	EV.	SOURCE
Thin porous liners are transparent at the blade tone	Quarter-wave: a 30 mm layer peaks near 2.8 kHz; 250 Hz needs 343 mm. Measured: 60 mm melamine and PET both stay below $\alpha = 0.3$ at 250 Hz. Compressing a thick batt into a thin cavity destroys its low-frequency absorption.	T	Koszałka materials review, pp. 2–3
Causality sets the floor	No passive material can be flat-broadband from ~400 Hz at 10–30 mm; narrowband resonators tuned to one harmonic are exempt.	T	Koszałka materials review, pp. 2, 8

RULE	NUMBER AND CONDITION	EV.	SOURCE
Thin resonant liners do cut real fan tones	23 mm Helmholtz metamaterial on a 120 mm 7-blade fan (BPF 605 Hz), measured: $-6.86 / -10.76 / -8.98 / -4.46$ dB(A) at the first four harmonics, mass flow -2.6% . Four units were fitted, one per duct wall; the authors warn that a single unit may not be enough in a realistic duct.	E	Wu et al. 2025, pp. 3, 5–6, 9
... including under real flow	15 mm MPP + membrane liner in a server fan: -3 dBA at the 620 Hz blade tone, -2.5 dBA overall.	E	Killeen 2023, pp. 1–2
Broader bands cost depth	Graded cascade of 9 resonators: quasi-perfect absorption 300–1000 Hz at 11 cm. Coiled cavities behind an MPP: mean α 0.91 over 400–1600 Hz at < 65 mm. Both impedance-tube only.	E	Jiménez 2017, p. 1; Min 2025, p. 1
Over-the-tip liners work, but not at every harmonic	Propeller duct with a metal-foam liner over the tip: up to -4 dB in-duct broadband over 1–5 BPF; on-axis far field -6 dB at BPF but almost $+9$ dB at $2\times$ BPF. On a large low-speed fan, the same idea gave 5 dB vs 3 dB for an inlet liner, and a tighter tip gap helped.	E	Pallejà-Cabré 2025, pp. 1, 7; Sutliff-Jones 2009, pp. 7, 9
Bury a tip liner in grooves	A flush porous surface over the tip cost 2.75–8.75 % fan efficiency; the same treatment at the bottom of circumferential grooves cost as little as 0.75 %. Turbofan-scale data; the geometry lesson is what transfers.	T	fan_noise_treatments_sota §5.2
Printed liners fail at the seams	A few scratches in the sealing tape cut the transmission-loss peak by 10 dB. A 0.6 mm rupture in a printed PET-G wall halved peak α . Micro-perforated facings are not an FDM process. Which face points into the duct shifts the absorption peak.	E/T	Jamois 2025, pp. 10, 15; Koszałka metamaterials review, pp. 5–6
Vendor absorption curves read high	Thinsulate data are random-incidence (Alpha Cabin), typically 0.1–0.3 above normal-incidence tube values for the same specimen.	T	Koszałka materials review, p. 4
Enclosure materials: cork and rubber help only above 1 kHz	Cork $\alpha \approx 0.02$ at 100 Hz (20 mm), > 0.6 over 1–5 kHz; recycled-rubber composites best as an impact-resistant outer skin. No source found measures EPP, XPS, cardboard or felt.	T	Koszałka materials review, p. 6

05 Beyond the propeller

RULE	NUMBER AND CONDITION	EV.	SOURCE
The motor can be the noise at low rpm	Coaxial rig: motor dominates at 1140–1500 rpm, propeller above ~ 1860 rpm. On a two-stroke UAV the engine dominates outright.	E	Russo 2023, p. 1; Moshkov 2021, p. 2
Periodic errors make tones, random errors make hiss	A repeatable rpm error or blade vibration adds discrete tones; a random one raises the broadband floor. Balance and speed control are acoustic design items.	S	Zhong 2023, p. 1
Inflow turbulence matters to listeners more than to meters	More inflow turbulence left equivalent SPL nearly unchanged but raised broadband, loudness and psychoacoustic annoyance.	E	Merino-Martínez 2025, p. 1
Design for perception	Optimised propeller: level change < 1.8 dB, psychoacoustic annoyance -20% , loudness -3.6 sone, sharpness -0.47 acum. Annoyance is driven by perceived noise level and sharpness; shorter impulsive bursts are more annoying per dB (1 ms needs 6–8 dB less than 20 ms).	E	Merino-Martínez 2024, pp. 6–7; Torija 2022, p. 1; Schlittenlacher 2023, p. 5
Drones are judged harsher than their level	Systematic review: more annoying than road or aircraft noise at the same exposure. Telling listeners a drone was on a medical delivery lowered annoyance for the same sound.	E/R	Schäffer 2021, p. 1; Woodcock 2025, p. 1
Operations are a design lever too	WHO indoor sleep limits can be worked back through a façade into a minimum flight distance. An ISO 3744 sound-power figure predicted flyover levels within ± 2.1 dB up to 30 m.	S/E	Ramos-Romero 2022, p. 1; Cussen 2022, p. 1

06 What this means for our two research trees

These are our readings of the rules above against the questions in duct-research-tree and drone-paczek-research-tree. They are predictions to test, not results.

Duct tree (6-in propeller, $R \approx 76$ mm)

- **Tip gap 1 vs 3 mm** is 1.3 % vs 3.9 % of radius. That straddles the drop the 5-in CFD put between 1.1 % and 3.6 %. Expect the 3 mm duct to lose most of the thrust gain. The acoustic literature, thin as it is, says the tip gap matters less for noise than whether the lip flow separates.
- **Duct height 25/50/100 mm.** Length is aerodynamically second-order. The acoustic lever is centring the propeller axially, which a taller duct makes possible. Separately, a thickness of ≥ 15 mm at the prop plane (10 % of D) is the cited guard against static separation.
- **Rod count.** The only evidence is fenestron-scale: move struts away from the rotor plane (5.5 % D_t gave up to 10 dB). Rod count sets which rotor-strut tones appear (Tyler-Sofrin, per the team review).
- **Test static, compare at equal thrust.** A static test may well show the duct louder (Malgoezar), so report thrust with every capture.
- **Liner placement.** Inside at the tip is where a liner has the most effect, and where it costs the most thrust unless it is buried in grooves. 25 mm porous liners will not touch the 200–670 Hz blade tone; a resonant liner might.
- **Motor spacing.** This is side-by-side rotor spacing, which has no measured noise rule. A spacing sweep on our rig would be new data.

Parcel-drone tree (enclosure linings)

- No paper found measures **EPP, XPS, cardboard or felt** acoustically. Our dp1 captures are the only data on them we know of.
- At tens of millimetres, expect every porous lining to do little below ~ 500 Hz, and differences to show above ~ 1 kHz. Cork and Thinsulate literature agrees.
- If the propeller sweep finds a strong blade tone, only a tuned resonator in the lining can target it. Seams and leaks can cost 10 dB of the effect.
- Rank materials with loudness and sharpness as well as dB(A), since a lining that removes high-frequency hiss may matter more to listeners than its level change suggests.

On our own arc

Layout effects of 3–8 dB are within what the arc can resolve in broadband, but blade tones below ~ 1 kHz are still provisional on it, and the hub sphere is not axisymmetric at 3–5 kHz. Treat arc results on tones with the same caution as arc-rig-onepager.pdf.

07 Where the sources disagree, and what nobody has measured

- **Toroidal propellers:** a careful 5-in test found one louder by up to 10 dB; other studies report 5–20 dB reductions. Methods differ widely.
- **Cost of tripping:** a thin tape trip reported negligible loss; grit and rough-surface trips measured 3–11 % thrust. The trip type decides it.
- **Serrations:** clean reductions on machined blades, increases on printed cut-in blades. Say how it was made.
- **Diffuser angle:** no effect, a negative effect, and a positive-then-negative effect in three historical studies. Each held a different parameter fixed.
- **Tip speed:** at fixed blade shape, lower tip speed is quieter; across blade shapes, a quieter optimised blade once spun at twice the rpm (Sinibaldi 2012).
- **Coaxial optimum and size:** 0.2–0.4 D for most studies, 0.1 D for one 14-in rotor.
- **Not found anywhere:**
 - a measured noise rule for side-by-side rotor spacing on a vehicle with arms;
 - round vs rectangular vs aerofoil arm sections;
 - tandem-rotor spacing;
 - landing-gear noise on small multirotors;
 - leading-edge tubercles or coatings on small propellers;
 - any acoustic data on EPP, XPS, cardboard or felt;
 - a per-propeller vs whole-drone duct comparison with both thrust and noise.

08 Papers still to fetch (PW proxy or OneDrive)

These screened as relevant but were paywalled or blocked. No claim above rests on them. The most useful come first.

PAPER	IDENTIFIER	WOULD SETTLE
Candeloro, Zaghi & Ferro 2020, shrouded 5–6-in propellers (Drones)	no DOI in the citing review	tip-gap and tolerance trends at our propeller size
Guo, Zhou, Fang & Zhang 2021, compact lined duct for small propellers	Appl. Acoust. (vol. 179 or 184, disputed)	liner dB on top of duct shielding
Go, Kingan, Wu & Sharma 2023, shrouded UAV propeller sound field	Appl. Acoust. 211, 109523	shroud quieter downstream, louder upstream

PAPER	IDENTIFIER	WOULD SETTLE
Pallejà-Cabré et al. 2024, OTR liners on a shrouded propeller	AIAA 2024-3407	anchor paper of our liner design brief
Zhong, Yang & Åbom 2023, EDF over-the-rotor treatment	10.1016/j.apacoust.2023.109298	3–6 dB claim, still secondary
Yang et al. 2025, ultra-thin metamaterial on an 11-in ducted propeller	10.1016/j.ast.2025.111121	8 dB BPF cut at 20 mm
Hrishikeshavan, Black & Chopra 2014, quad-shrouded MAV	J. Aircraft 51(3), 779–791	shroud mass penalty at MAV scale
Reboul & Gallo 2024, shrouded vs open coaxial rotors	hal-04582952	shroud A/B on coaxial rotors
Jdiobe et al. 2025, open vs ducted counter-rotating UAV propellers	10.2514/1.J065433	computational A/B
Aleksandrov & Penkov 2012, optimal gap between quadrotor rotors	DAAAM Baltic 2012, pp. 251–255	the one paper titled on side-by-side spacing
Torija et al. 2021, psychoacoustics of contra-rotating UAV propellers	10.1121/10.0003432	primary for the z/D 0.2–0.4 annoyance optimum
Jardin et al. 2023, solidity and low-Re rotor noise	10.1177/1475472x221150181	blade count vs chord as routes to solidity
Turhan et al. 2024, phase-synchronised adjacent propellers	10.1121/10.0025990	24 dB in forward flow
Gan 2025, broadband modulation of multirotors	10.1121/10.0035646	phase and amplitude modulation on a hexacopter
Santamaría et al. 2024, serration shape	10.1016/j.jsv.2024.118771	square vs sawtooth vs sinusoidal teeth
Rong et al. 2025 / Wei et al. 2025, owl-inspired serrations	10.1063/5.0268150 · 10.1063/5.0266266	leading-edge serration cost and gain
Du Plessis & Bouferrouk 2024, looped blades	AIAA 2024-3156	in-plane vs out-of-plane loops
Kawai et al. 2024, drone vs transport annoyance	10.1121/10.0032386	annoyance at equal exposure
Trematerra & Lombardi 2017; Lara-Sáenz 2018	Key Eng. Mater. 744; Constr. Build. Mater.	cork and rubber absorption curves
Calliccoat 2016, composite skins for UAV transmission loss	SHAREOK handle 11244/48787	body-skin transmission loss, relevant to the parcel box

Full lists: the "Not retrieved" sections of the four EXTRACTS files (the duct file alone lists 20 rows) and SCREENING-BROAD.md (133 screened, 63 relevant, 34 fetched). The OneDrive is not mounted on this laptop, so anything held only there needs syncing first.

Method. On 2026-09-28, four threads extracted rules into papers/SoundVisualizer/design-rules/:

- layout: 94 quotes;
- propellers: 225 quotes;
- ducts and linings: 268 quotes;
- a web sweep through OpenAlex and arXiv: 68 quotes from 34 open-access PDFs.

Every quoted passage was matched verbatim against a pdftotext dump by scripts/verify_quotes.py, as were the quotes on this page. Page numbers are the PDF page index unless marked "printed". Two corrections were made against the source while writing this page:

- the Zawodny-Boyd clearance step is 0.1→0.2 R, not 0.1→0.3 R;
- the team ducting review's "airframe modifications alone" result is in fact rotor elevation on standoffs.

Evidence built on the team's own reviews [T] is only as strong as their sources; where those sources were read here, they are cited directly instead. This is a quick scan, not a systematic review.